

REST-API-Endpoint-Design

Purpose

This document defines the REST API endpoints for the Employee Management & Payroll System. It organizes endpoints by functional module and specifies the HTTP method, operation, and role-based access for each endpoint. The endpoints are designed to be implemented using FastAPI and interact with the system's SQL Server database.

A module is not based on a particular role or database table. Modules are created to group and organize specific tasks related to a particular functional area of the system.

Documentation Ground Rules

1. Use consistent endpoint naming
 - Use plural resource names: /employees, /departments, /payrolls.
 - Use lowercase and hyphens where needed.
 - Keep the /api/ prefix consistent.
2. Follow HTTP method conventions
 - GET → Retrieve data
 - POST → Create data / perform an operation
 - PUT → Update an existing resource
 - PATCH → Partial update or status/action change
 - DELETE → Remove a resource
3. Separate modules from actions
 - A module represents a functional area of the system.
 - Actions represent operations that can be performed within that module.
4. Define role permissions separately
 - A module itself does not belong to a particular role.
 - Individual actions/endpoints are restricted according to the user's role.
5. Use resource IDs where required
 - Example: /api/employees/{employee_id}
 - This identifies a specific resource.
6. Keep authentication and authorization separate
 - Authentication determines who the user is.
 - Authorization determines what that user is allowed to do.
7. Document the purpose of every endpoint

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- Each endpoint should clearly state what operation it performs and which roles can access it.

Modules & their actions

Authentication

Method	Endpoint	Action
POST	/api/auth/login	Login
POST	/api/auth/logout	Logout
POST	/api/auth/change-password	Change password

Employees

Method	Endpoint	Action
GET	/api/employees	Get all employees
GET	/api/employees/{id}	Get employee by ID
POST	/api/employees	Create employee
PUT	/api/employees/{id}	Update employee
DELETE	/api/employees/{id}	Delete employee

Departments

Method	Endpoint	Action
GET	/api/departments	Get all departments
GET	/api/departments/{id}	Get department by ID
POST	/api/departments	Create department
PUT	/api/departments/{id}	Update department
DELETE	/api/departments/{id}	Delete department

Designations

Method	Endpoint	Action
GET	/api/designations	Get all designations
GET	/api/designations/{id}	Get designation by ID
POST	/api/designations	Create designation
PUT	/api/designations/{id}	Update designation
DELETE	/api/designations/{id}	Delete designation

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Attendance

Method	Endpoint	Action
GET	/api/attendance	Get attendance records
GET	/api/attendance/{id}	Get attendance by ID
GET	/api/employees/{id}/attendance	Get employee attendance
POST	/api/attendance	Mark attendance
PUT	/api/attendance/{id}	Update attendance
DELETE	/api/attendance/{id}	Delete attendance

Leave

Method	Endpoint	Action
GET	/api/leaves	Get leave records
GET	/api/leaves/{id}	Get leave by ID
POST	/api/leaves	Apply for leave
PUT	/api/leaves/{id}	Update leave
DELETE	/api/leaves/{id}	Cancel/delete leave
PATCH	/api/leaves/{id}/approve	Approve leave
PATCH	/api/leaves/{id}/reject	Reject leave

Salary Structure

Method	Endpoint	Action
GET	/api/salary-structures	Get all salary structures
GET	/api/salary-structures/{id}	Get salary structure
POST	/api/salary-structures	Create salary structure
PUT	/api/salary-structures/{id}	Update salary structure
DELETE	/api/salary-structures/{id}	Delete salary structure

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Payroll

Method	Endpoint	Action
GET	/api/payrolls	Get payroll records
GET	/api/payrolls/{id}	Get payroll by ID
GET	/api/employees/{id}/payrolls	Get employee payroll history
POST	/api/payrolls	Generate payroll
PUT	/api/payrolls/{id}	Update payroll
DELETE	/api/payrolls/{id}	Delete payroll

Payroll Details

Method	Endpoint	Action
POST	/api/payrolls/{id}/details	Add payroll details
PUT	/api/payroll-details/{id}	Update payroll details
DELETE	/api/payroll-details/{id}	Delete payroll details